

HAMZA JEDDAD

Senior AI / ML Engineer

@ jeddad.hamza@gmail.com

+33 6 19 47 01 79

in <https://www.linkedin.com/in/hjeddad/>

Paris, France

PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 6+ years of experience specializing in **conversational AI**, **enterprise chatbots**, and **intelligent automation**. Proven track record building **production chatbots with RAG**, **LLM orchestration**, and **enterprise system integration**. Expert in **GDPR-compliant AI systems**, IT helpdesk automation, and reducing repetitive tasks through intelligent workflows. Strong experience with **LangChain/LangGraph**, **knowledge base integration**, **ticketing systems**, and **full remote collaboration**.

WORK EXPERIENCE

Senior AI Engineer

Leboncoin

Nov 2025 – Feb 2026

Paris, France

- Contributed to the development and optimization of **Markus**, Leboncoin's customer support chatbot, serving millions of users across France's leading classifieds platform.
- Designed and implemented **conversational AI flows** using **LLM-based dialogue management**, integrating intent classification, entity extraction, and context-aware response generation.
- Built **RAG (Retrieval-Augmented Generation)** pipelines to ground chatbot responses in Leboncoin's knowledge base, FAQs, and policy documents, reducing hallucinations and improving answer accuracy.
- Implemented **guardrails and safety mechanisms** to handle edge cases, inappropriate queries, and escalation to human agents when needed.
- Architected **integration patterns** for connecting chatbot to backend systems, knowledge bases, and ticketing workflows, ensuring **GDPR compliance** and data sovereignty.
- Designed **autonomous triage and escalation logic** to minimize manual intervention while maintaining quality, reducing IT team workload by intelligently routing complex cases.
- Collaborated with product, UX, and customer support teams to iterate on conversation flows based on user feedback and support ticket analysis.
- Monitored chatbot performance metrics (resolution rate, user satisfaction, escalation rate) and implemented continuous improvements based on production data.

Stack: Python, LangChain, LangGraph, OpenAI/Anthropic APIs, RAG, FastAPI, Docker, AWS/GCP

Senior AI Engineer

Mérieux NutriSciences

Oct 2025 – Feb 2026

Full Remote

- Mapped heterogeneous inputs across **cities**, **labs**, **parameters (chemistry & biology)**, and **status** to canonical values in **one reference table per dimension**.
- Built a matching engine combining **rules + fuzzy matching** with **LLM-assisted mapping (Gemini with tool/function calling)** for candidate generation and validation; applied confidence scores and thresholds.
- Implemented **human-in-the-loop** review: results routed to reviewers for approval/correction; captured feedback for continuous improvement and auditability.
- Engineered ingestion, validation, and export tools to support repeatable batch jobs and future real-time endpoints.

Stack: Python, Gemini (function calling); AWS

Senior AI/ML Engineer

Interencheres

Nov 2024 – Oct 2025

Paris, France

- Built production AI systems for catalog enrichment and pricing across **3M items/year**. Collaborated daily with business experts (métier) and backend engineers; added **human-in-the-loop** validation to all LLM services.
- AI Classification (hierarchical, multi-label)**
 - Designed a tree-based taxonomy (domain → subdomain → category → subcategory), expanding from **63** → **108** categories; supported **up to 3 labels** per domain.
 - Hierarchical prompting with **LangChain/LangGraph**; each node returns **Pydantic-validated JSON** (label, confidence, justification). Guardrails: schema checks, strict node lists, and context minimization to reduce hallucinations.

- Built a **Flask annotation platform** for sampling-based review (by seller/period/region); tracked experiments and metrics in **MLflow**. Added **seller post-prod validation** to measure live accuracy.
- Benchmarked multiple providers (OpenAI mini, Bedrock Titan/Llama, Gemini) for latency/cost/quality; selected **OpenAI mini** as best trade-off.
- **Outcome: 95% accuracy** on validation; powering category filters on site.
- **Category-Aware Characteristics Extraction**
 - From raw descriptions/images, extracted **category-dependent attributes** (e.g., cars: brand/model/mileage; jewelry: material/weight) to unlock **filters, dashboards, similar-item retrieval**.
 - Prompt-based extraction with **strict JSON schema**, multilingual models, **unit normalization**, and **ReAct-style** reasoning. Required **evidence/justification** for each field to limit hallucinations.
 - Same HIL workflow via annotation UI; deployed as **FastAPI** on AWS with Docker; shared observability via MLflow.
 - **Impact:** New filter facets enabled across key categories; quantitative UX gains under measurement.
- **Image-Driven Item Valuation (advanced prototype)**
 - Pipeline: **DINOv2-Large** embeddings → **Qdrant** vector search (visual comps) → LLM **image description** → **classification** → **Elasticsearch more_like_this** (+ category filter) → LLM **similarity judge & query refinement** → LLM **price estimation** from comparables.
 - Ground truth from historical **hammer prices**; achieved **30% average error** on test. Focused on improving retrieval under noisy descriptions via multi-step query expansion.
 - Status: experimental; integrating feedback loops and cost/latency controls prior to production.
- Infra: **FastAPI**, Docker, **AWS EC2/SageMaker**, asyncio; tooling: **LangChain**, **LangGraph**, **MLflow**, **Pydantic**, Qdrant, Elasticsearch; LLMs: OpenAI mini, Bedrock (Titan/Llama), Gemini.

Stack: Python, LangChain, LangGraph, MLflow, Pydantic, FastAPI, Flask, Docker, AWS EC2/SageMaker, Qdrant, Elasticsearch, OpenAI/Bedrock/Gemini, asyncio

Senior ML Engineer

AntVoice

📅 Nov 2022 – Oct 2024

📍 Paris, France

- Contributed to AntVoice's DSP managing ad campaigns on the open web (RTB). Operated at **1B+ queries/day** with strict latency and budget constraints.
- **RTB Prediction Models (CTR/CVR/Viewability/Qualification)**
 - Built models for post-impression and post-view clicks, conversions, view rate, and post-click qualification.
 - Designed the bid pricing function ($price = f(p_{click}, p_{view}, p_{conv}, \dots)$) integrating market dynamics, win-rate, and margin constraints under fixed monthly budgets.
 - Led feature engineering & audience segmentation (retargeting vs. acquisition). **Result: +20% CTR** (1.0 → 1.2%).
 - Evaluation: offline (AUC, logloss, calibration, precision/recall); online (incremental lift vs. control), using instant feedback loops (click/visit/conv).
 - Challenges: **data drift** (automated retraining & alerts), **latency <20ms** (lightweight/linear models, ONNX), and **big data** processing.
- **A/B Testing & Experimentation**
 - User-level randomized traffic split (A/B/C) to challenge new models fairly against baseline; near real-time monitoring in **Looker Studio**.
 - Promotion criteria: **CTR + margin + robustness over ≥ 2 days**. Prevented regressions versus the previous "deploy and hope" approach.
- **MLOps & Monitoring**
 - Orchestrated pipelines with **Airflow**; containerized deployments to **Vertex AI**. Monitoring for feature drift, unseen values, performance degradation; threshold-based retraining; **human-in-the-loop** approval for prod.
- **Collaboration & Reporting**
 - Weekly sessions with account managers; monthly product updates to cross-functional stakeholders.

Stack: Python, Scikit-learn, CatBoost, ONNX, Airflow, Vertex AI, BigQuery, Looker Studio, Docker, GCP

Data Scientist

Direct Assurance (AXA Group)

📅 Jan 2020 – Oct 2022

📍 Paris, France

- Worked across two departments: **Claims** (automation) and **Actuarial** (commercial premium optimization).
- **Claims Automation**
 - Built OCR+NLP extraction for diverse documents (invoices, records, forms): vendor, amounts, tax, dates, references; added rule-based validation.
 - Implemented **RPA (Selenium)** and overnight batch jobs to automate data entry into back-office systems.
 - **Impact:** Replaced fully manual processes, reducing lead time and errors for routine claims.
- **Commercial Premium Optimization (Actuarial)**

- For new car prospects: inputs (age, address, car details) feed **conversion models** (LogReg, CatBoost) designed to capture **price elasticity**.
- Combined **conversion**, **pure premium**, and **renewal** models to forecast **7-year CLV** and expected profitability.
- Two-step optimization: (1) simulate price scenarios ($\pm 30\%$ around pure premium) under constraints (loss ratio, turnover, acquisition volume); (2) train a **meta-model** mapping features $\rightarrow$ optimal margin.
- Cadence: new releases every **2 months**, with typical **+0.5pt** gains per iteration. Delivered Dash/Plotly dashboards and Databricks notebooks (PySpark) for A/B testing.
- Collaboration: actuaries (assumptions incl. inflation), data engineers (prod), and business teams; field discovery with métier for claims.

Stack: Python, SQL, Azure, Databricks, PySpark, CatBoost, Scikit-learn, Dash, Plotly, Selenium

NLP Research Intern

AXA France

📅 May 2019 – Oct 2019

📍 Paris, France

- Fine-tuned BERT-based NER models for unstructured insurance documents (beneficiary, claim type, coverage) using transfer learning.
- Improved extraction quality through domain-specific tokenization and careful hyperparameter tuning.

Tools: BERT, Transformers, PyTorch, TensorFlow

Developer & Data Science Intern

Paris Digital Lab

📅 Feb 2017 – Jul 2017

📍 Paris, France

- Built recommendation systems and chatbots for clients (AccorHotels, Foncia) to automate customer support and engagement.
- Managed delivery from requirements to working prototypes.

Tools: Python, NLP, Machine Learning, Pandas, Scikit-learn

EDUCATION

Master of Science: Artificial Intelligence

CentraleSupélec (École Centrale Paris)

📅 Sept 2018 – Dec 2019

📍 Paris, France

Master of Engineering: Applied Mathematics & Computer Science

CentraleSupélec (École Centrale Paris)

📅 Sept 2015 – Dec 2019

📍 Paris, France

SKILLS

- **GenAI & LLMs:** Conversational AI, enterprise chatbots, LangChain, LangGraph, RAG, ReAct, prompt engineering, fine-tuning, multimodal flows, guardrails, LangSmith/MLflow.
- **Enterprise Integration:** Teams, SharePoint, ticketing systems, GDPR compliance, data sovereignty, IT automation.
- **Machine Learning:** Scikit-learn, CatBoost, XGBoost, PyTorch, TensorFlow, ONNX.
- **Data Engineering:** BigQuery, PostgreSQL, MongoDB, Elasticsearch, Qdrant, Airflow, Spark.
- **MLOps:** Vertex AI, SageMaker, Docker, CI/CD, observability, monitoring, drift detection.
- **Software Engineering:** Python, FastAPI, asyncio, Flask, Dash, Git, testing, code reviews.
- **Cloud:** AWS, GCP, Azure.
- **Visualization:** Dash, Plotly, Looker Studio.

LANGUAGES

English: Professional proficiency

French: Bilingual

Arabic: Native